

A Semi-Autonomous SIEM–SOAR Framework for Anomaly-Based Security Operations with Human-in-the-Loop Learning

A Major Project Report Submitted by

G.G. Venkateswararao (Reg. No. 221FA19020)

G. S. N. S. Niteesh Babu (Reg. No. 221FA19030)

V. Tarun Kumar Reddy (Reg. No. 221FA19057)

Saugat Chaudhary (Reg. No. 221FA19066)

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BACHELOR OF TECHNOLOGY

in

Cyber Security

Under the Guidance of

Dr. Benson Mansingh P.M



Department of Advanced Computer Science and Engineering

School of Computing and Informatics

Vignan's Foundation for Science, Technology & Research

(Deemed to be University)

Andhra Pradesh–522213, India

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Department of Advanced Computer Science and Engineering
School of Computing and Informatics
Vignan's Foundation for Science, Technology & Research
(Deemed to be University)
Andhra Pradesh, India-522213

CERTIFICATE

This is to certify that the project entitled **“A SEMI-AUTONOMOUS SIEM–SOAR FRAMEWORK FOR ANOMALY-BASED SECURITY OPERATIONS WITH HUMAN-IN-THE-LOOP LEARNING”** has been carried out by **G. Gangadara Venkateswararao (221FA19020)**, **G. S. N. S. Niteesh Babu (221FA19030)**, **V. Tarun Kumar Reddy (221FA19057)**, and **Saugat Chaudhary (221FA19066)**, in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology in Cyber Security**.

The work was carried out in the Department of Advanced Computer Science and Engineering, Vignan's Foundation for Science, Technology and Research (Deemed to be University), Vadlamudi, Guntur District, Andhra Pradesh, India, under my guidance and supervision. This project was successfully completed during the AY 2025–26.

Supervisor

Project Co-ordinator

HOD, ACSE

Dr. Benson Mansingh
P.M

Dr. Amar Jukuntla

Dr. D Radha Rani

External Examiner



Department of Advanced Computer Science and Engineering
School of Computing and Informatics
Vignan's Foundation for Science, Technology & Research
(Deemed to be University)
Andhra Pradesh, India-522213

DECLARATION

We hereby declare that our project work described in the project titled **“A SEMI-AUTONOMOUS SIEM–SOAR FRAMEWORK FOR ANOMALY-BASED SECURITY OPERATIONS WITH HUMAN-IN-THE-LOOP LEARNING”** which is being submitted by us for the partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in the Department of Advanced Computer Science and Engineering, Vignan's Foundation for Science, Technology and Research (Deemed to be University), Vadlamudi, Guntur, Andhra Pradesh, is a record of the work carried out by us under the guidance of **Dr. Benson Mansingh P.M..**

G.G. VENKATESWARARAO
(221FA19020)

G.S.N.S. NITEESH BABU
(221FA19030)

V. TARUN KUMAR REDDY
(221FA19057)

SAUGAT CHAUDHARY
(221FA19066)

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ABSTRACT

Security Operations Centers (SOCs) monitor large volumes of security events generated by enterprise networks, cloud platforms, and authentication systems. Traditional rule-based SIEM systems effectively detect known threats but often fail to identify sophisticated attacks such as insider threats and zero-day intrusions while generating excessive alerts. Fully autonomous response systems may also introduce operational risks without analyst supervision. This project presents a semi-autonomous SIEM–SOAR framework integrated with anomaly detection and human-in-the-loop learning. The framework combines machine learning-based anomaly detection, centralized SIEM monitoring, guided SOAR automation, and analyst-assisted validation to improve cybersecurity operations while maintaining analyst control. The system processes CICIDS-style network traffic and SSH authentication logs using preprocessing and behavioral feature extraction techniques. An Isolation Forest model identifies anomalous activities, and detected alerts are visualized through ELK Stack and Kibana dashboards. High-severity alerts are forwarded to the Shuffle SOAR platform for notification and case tracking using webhook-based workflows. Analyst feedback is incorporated through periodic retraining to improve future detection performance and reduce false positives. Experimental results demonstrate that the proposed framework effectively improves anomaly detection, reduces alert overload, and enhances operational efficiency, providing a scalable solution for intelligent security operations.

Keywords: SIEM, SOAR, Security Operations Center, Anomaly Detection, Isolation Forest, Human-in-the-Loop Learning, ELK Stack, SOC Automation, Cybersecurity, Machine Learning.

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List of Acronyms/Abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
CSV	Comma Separated Values
CICIDS	Canadian Institute for Cybersecurity Intrusion Detection System
ELK	Elasticsearch, Logstash, and Kibana
IDS	Intrusion Detection System
ML	Machine Learning
SIEM	Security Information and Event Management
SOAR	Security Orchestration, Automation, and Response
SOC	Security Operations Center

List of Symbols

X	Input security telemetry dataset
x_i	Individual security event
$f(x)$	Isolation Forest scoring function
$S(x)$	Anomaly score
T	Detection threshold value
N	Number of security events
μ	Mean value of features
σ	Standard deviation

Chapter 1

Introduction

This chapter introduces a semi-autonomous SIEM–SOAR framework for anomaly-based cybersecurity operations with human-in-the-loop learning. It covers SOC background, limitations of traditional systems, motivation, problem statement, and report structure.

1.1 Background

Modern organizations depend on cloud systems, distributed applications, enterprise services, and interconnected networks that continuously generate massive volumes of security-related data. This includes firewall logs, authentication events, DNS queries, endpoint telemetry, application logs, and network flow records. As digital transformation accelerates, the complexity and scale of cybersecurity monitoring have increased significantly.

Security Operations Centers (SOCs) are dedicated units responsible for monitoring, detecting, analyzing, and responding to cybersecurity incidents in real time. SOC analysts rely on multiple tools to correlate events and identify potential threats across heterogeneous data sources. The primary objective of a SOC is to ensure confidentiality, integrity, and availability of organizational assets while minimizing the impact of cyber incidents.

Traditional SOC environments are heavily dependent on Security Information and Event Management (SIEM) systems. SIEM platforms aggregate logs from multiple sources into a centralized repository and provide correlation rules, dashboards, and alert generation mechanisms. They play a critical role in incident detection and forensic investigation.

However, conventional SIEM systems are fundamentally rule-driven. They depend on predefined signatures and static correlation logic to identify malicious activity. While effective for known attack patterns, they are inadequate for detecting unknown or evolving threats such as advanced persistent threats (APTs), insider misuse, polymorphic malware, and zero-day exploits.

Modern attackers deliberately design techniques that evade signature-based detec-

tion. These attacks often appear as normal behavior over extended periods, making detection extremely difficult. As a result, SOC environments experience a high volume of alerts, many of which are false positives. This leads to alert fatigue, reduced analyst efficiency, and delayed incident response.

To address these challenges, machine learning-based anomaly detection techniques have been widely explored. Unlike rule-based systems, anomaly detection models learn patterns of normal behavior and identify deviations from expected system activity. This makes them suitable for detecting previously unseen attacks.

Among various algorithms, Isolation Forest is widely used due to its simplicity, scalability, and effectiveness in high-dimensional datasets. It works on the principle of isolating anomalies rather than profiling normal behavior. Since anomalies are rare and different, they require fewer partitions to isolate, making detection computationally efficient.

In addition to detection mechanisms, Security Orchestration, Automation, and Response (SOAR) platforms enhance SOC efficiency by automating repetitive tasks such as alert enrichment, ticket creation, notification handling, and workflow orchestration. However, full automation can introduce operational risks if incorrect actions are executed without human validation.

Therefore, modern cybersecurity systems require a balanced approach that integrates automation with human oversight. The proposed framework introduces a semi-autonomous SIEM-SOAR architecture that combines machine learning-based anomaly detection, centralized SIEM visualization, SOAR-driven automation, and human-in-the-loop feedback learning.

The system processes CICIDS-style network datasets along with synthetic SSH authentication logs. These events are normalized and structured into behavioral windows for anomaly detection. Detected anomalies are stored in the ELK Stack, enabling visualization and investigation through Kibana dashboards. Critical alerts are forwarded to SOAR workflows for response handling while maintaining analyst control over final decisions. Feedback from analysts is used to continuously improve detection accuracy through retraining.

Modern enterprise environments are rapidly evolving due to cloud adoption, IoT integration, remote working models, and distributed computing systems. These developments have significantly expanded the attack surface, allowing adversaries to exploit multiple entry points within enterprise networks.

Cyber attacks today are highly sophisticated, often combining multiple techniques such as credential theft, lateral movement, privilege escalation, and persistence mechanisms. These attacks are designed to remain undetected for long durations, making

them difficult for traditional security systems to identify.

SOC environments face another critical challenge in the form of alert overload. Large enterprises generate millions of log events daily, resulting in thousands of alerts that require investigation. Most of these alerts are benign or redundant, leading to inefficiencies in manual analysis.

Machine learning-based anomaly detection helps mitigate this issue by identifying behavioral deviations rather than relying on fixed rules. This significantly improves detection capability for unknown and emerging threats.

The proposed framework integrates detection, visualization, automation, and feedback mechanisms into a unified architecture, making it suitable for real-world SOC environments.

1.2 Motivation for the present research work

The increasing complexity of modern cybersecurity environments necessitates intelligent, scalable, and adaptive security solutions. Traditional SOC systems are not designed to handle the volume, velocity, and variety of modern security telemetry.

Organizations generate vast amounts of data from firewalls, endpoints, authentication systems, cloud platforms, and network monitoring tools. Manual analysis of such data is impractical and inefficient.

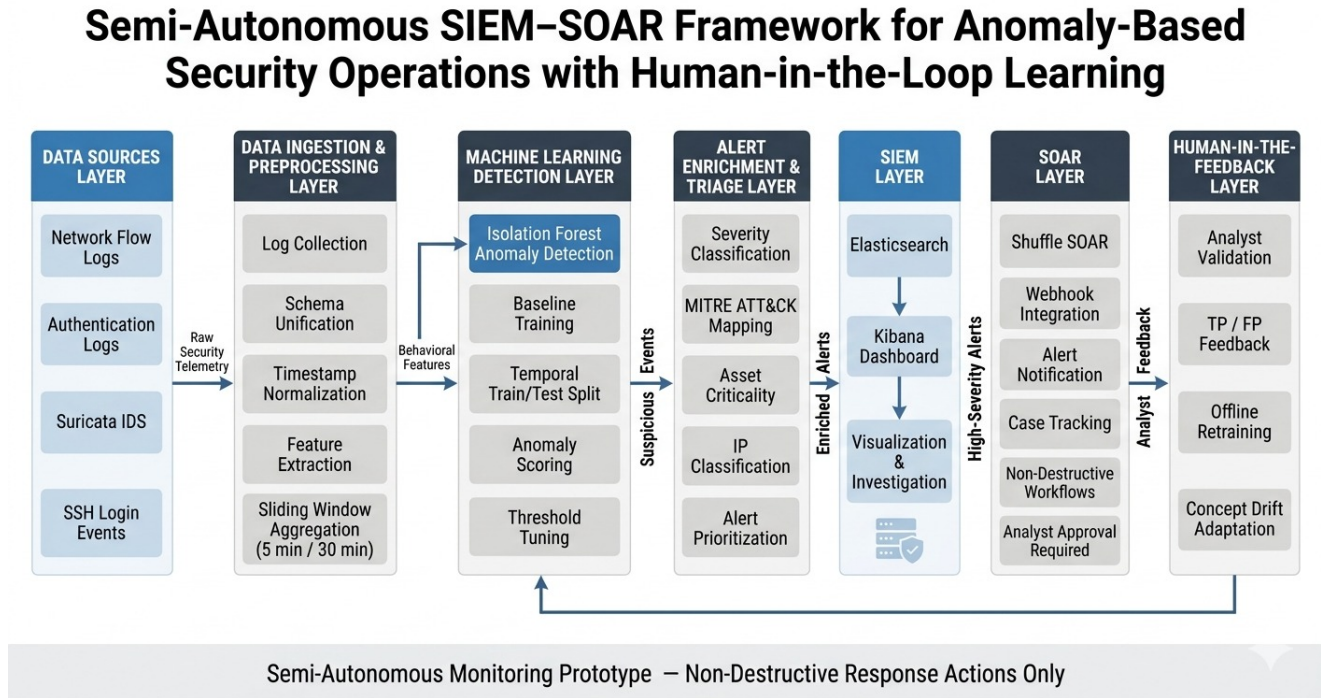


Figure 1.1: Semi-autonomous SIEM-SOAR framework architecture

Conventional SIEM systems depend heavily on static rules and signature-based detection methods. While effective for known threats, they fail to detect advanced and evolving attacks that do not follow predefined patterns.

Attackers increasingly use stealth-based techniques such as slow reconnaissance, lateral movement, and credential misuse to bypass detection systems. These attacks are difficult to detect using traditional rule-based approaches.

SOC analysts also face alert fatigue due to high volumes of false-positive alerts generated by rigid correlation rules. This reduces operational efficiency and increases the risk of missing critical incidents.

SOAR platforms improve efficiency by automating repetitive tasks; however, fully autonomous systems may introduce risks if incorrect actions are executed without validation.

The key motivations of this research include:

- Enhancing detection capability using machine learning models.
- Reducing analyst workload through intelligent automation.
- Improving visibility through centralized SIEM dashboards.
- Ensuring safe SOAR-based workflow execution.
- Maintaining human oversight in critical decisions.
- Supporting continuous learning using feedback loops.
- Developing a scalable SOC-ready architecture.

1.3 Problem statement

Traditional SIEM systems rely on rule-based and signature-based detection mechanisms that are insufficient for identifying unknown and evolving cyber threats.

Modern cyber attacks such as advanced persistent threats, insider attacks, and zero-day exploits often bypass detection systems due to their stealthy and adaptive behavior.

A major challenge in SOC environments is excessive false-positive generation, which leads to alert fatigue and reduces analyst efficiency. Analysts must manually investigate large volumes of alerts, many of which are not malicious.

Although SOAR systems automate incident response processes, fully autonomous actions may cause operational disruptions if incorrect decisions are executed.


```
saugat@saugat-virtual-machine: ~  
saugat@saugat-virtual-machine:~$ sudo suricata -T -c /etc/suricata/suricata.yaml -v  
11/1/2026 -- 12:25:13 - <Info> - Running suricata under test mode  
11/1/2026 -- 12:25:13 - <Notice> - This is Suricata version 6.0.4 RELEASE running in SYSTEM mode  
11/1/2026 -- 12:25:13 - <Info> - CPUs/cores online: 4  
11/1/2026 -- 12:25:14 - <Info> - fast output device (regular) initialized: fast.log  
11/1/2026 -- 12:25:14 - <Info> - eve-log output device (regular) initialized: /var/log/suricata/eve.json  
11/1/2026 -- 12:25:14 - <Info> - stats output device (regular) initialized: stats.log  
11/1/2026 -- 12:25:30 - <Info> - 1 rule files processed. 47461 rules successfully loaded, 0 rules failed  
11/1/2026 -- 12:25:30 - <Info> - Threshold config parsed: 0 rule(s) found  
11/1/2026 -- 12:25:31 - <Info> - 47464 signatures processed. 1251 are IP-only rules, 5180 are inspecting packet payload, 40816 inspect application layer, 108 are decoder e  
vent only  
11/1/2026 -- 12:27:26 - <Notice> - Configuration provided was successfully loaded. Exiting.  
11/1/2026 -- 12:27:27 - <Info> - cleaning up signature grouping structure... complete  
saugat@saugat-virtual-machine:~$ ping -c 1 testnids.com  
PING testnids.com (217.160.0.187) 56(84) bytes of data.  
64 bytes from 217-160-0-187.elastic-ssl.ul-r.com (217.160.0.187): icmp_seq=1 ttl=128 time=153 ms  
--- testnids.com ping statistics ---  
1 packets transmitted, 1 received, 0% packet loss, time 0ms  
rtt min/avg/max/ndev = 152.528/152.528/152.528/0.000 ms  
saugat@saugat-virtual-machine:~$ sudo suricata -T -c /etc/suricata/suricata.yaml -v  
11/1/2026 -- 12:28:29 - <Info> - Running suricata under test mode  
11/1/2026 -- 12:28:29 - <Notice> - This is Suricata version 6.0.4 RELEASE running in SYSTEM mode  
11/1/2026 -- 12:28:29 - <Info> - CPUs/cores online: 4  
11/1/2026 -- 12:28:29 - <Info> - fast output device (regular) initialized: fast.log  
11/1/2026 -- 12:28:29 - <Info> - eve-log output device (regular) initialized: /var/log/suricata/eve.json  
11/1/2026 -- 12:28:29 - <Info> - stats output device (regular) initialized: stats.log  
11/1/2026 -- 12:28:44 - <Info> - 1 rule files processed. 47461 rules successfully loaded, 0 rules failed  
11/1/2026 -- 12:28:44 - <Info> - Threshold config parsed: 0 rule(s) found  
11/1/2026 -- 12:28:45 - <Info> - 47464 signatures processed. 1251 are IP-only rules, 5180 are inspecting packet payload, 40816 inspect application layer, 108 are decoder e  
vent only  
11/1/2026 -- 12:30:33 - <Notice> - Configuration provided was successfully loaded. Exiting.  
11/1/2026 -- 12:30:33 - <Info> - cleaning up signature grouping structure... complete  
saugat@saugat-virtual-machine:~$
```

Figure 1.2: Suricata IDS network traffic monitoring and event logging

Existing systems also lack adaptive learning capabilities that incorporate analyst feedback to improve future detection performance.

Therefore, there is a need for a cybersecurity framework that:

- Detects anomalies using intelligent machine learning techniques.
- Reduces alert overload through prioritization.
- Provides centralized monitoring and visualization.
- Maintains human supervision in decision-making.
- Enables continuous improvement through feedback-based learning.

The proposed semi-autonomous SIEM–SOAR framework addresses these challenges by integrating anomaly detection, SIEM visualization, SOAR automation, and human-in-the-loop learning into a unified system.

The system processes network traffic and authentication logs derived from CICIDS datasets and SSH login simulations. These logs are normalized and grouped into behavioral time windows for analysis.

Isolation Forest is used for anomaly detection due to its efficiency in identifying rare and abnormal patterns in high-dimensional data.

Detected anomalies are stored in the ELK Stack, enabling visualization and investigation through Kibana dashboards.

High-severity alerts are forwarded to SOAR workflows for automated response handling, while maintaining analyst control. Feedback from analysts is used to retrain models and improve detection accuracy over time.

1.4 Organization of the project report

The report is structured into seven chapters:

- i)** Chapter 1: Introduction, background, motivation, and problem statement.
- ii)** Chapter 2: Literature review of IDS, SIEM, SOAR, and machine learning techniques.
- iii)** Chapter 3: Proposed methodology and system architecture.
- iv)** Chapter 4: Implementation details and experimental results.
- v)** Chapter 5: Conclusion and future scope.
- vi)** Chapter 6: Implementation files and setup procedures.
- vii)** Chapter 7: SDG mapping and impact analysis.

Chapter 2

Literature review

This Chapter presents a survey of the most commonly used cybersecurity monitoring systems, intrusion detection techniques, SIEM architectures, SOAR technologies, anomaly detection methods, and machine learning approaches used in modern Security Operations Centers (SOCs). The chapter also discusses existing research works related to anomaly-based threat detection, automated security operations, and human-in-the-loop cybersecurity systems.

2.1 Introduction to cybersecurity monitoring systems

Cybersecurity monitoring systems play an essential role in protecting enterprise infrastructures from malicious activities, unauthorized access, data breaches, malware infections, and network intrusions. Modern organizations continuously generate enormous volumes of security telemetry from firewalls, routers, servers, endpoint devices, authentication systems, cloud services, and communication networks. Monitoring and analyzing this large-scale telemetry manually is extremely difficult for security analysts.

Traditional cybersecurity systems primarily rely on rule-based detection and signature matching techniques to identify malicious activities. These systems compare incoming events against predefined attack signatures and correlation rules in order to generate alerts. Although signature-based systems are highly effective for detecting known threats, they often fail to identify sophisticated attacks that do not match existing signatures.

The increasing complexity of enterprise infrastructures has significantly increased the need for intelligent monitoring systems capable of identifying suspicious behavioral deviations in real time. Security Operations Centers (SOCs) therefore utilize advanced monitoring technologies including SIEM systems, intrusion detection systems, anomaly detection platforms, and automated response mechanisms to improve cybersecurity operations.

Anomaly detection has emerged as one of the most important research areas in

cybersecurity because of its capability to identify previously unknown attacks. Unlike signature-based approaches, anomaly detection systems learn normal system behavior and identify deviations from established behavioral baselines.

Machine learning-based cybersecurity monitoring systems provide adaptive detection capability and improved scalability for large-scale enterprise environments. These systems are increasingly integrated with Security Information and Event Management (SIEM) and Security Orchestration, Automation, and Response (SOAR) platforms to improve operational efficiency.

$$A(x) = N(x) + \eta(x) \quad (2.1.1)$$

where $A(x)$ represents observed activity, $N(x)$ represents normal behavioral activity, and $\eta(x)$ represents anomalous deviations detected within the monitored system.

2.1.1 Security Information and Event Management (SIEM)

Security Information and Event Management (SIEM) systems are widely used in modern SOC environments for centralized security monitoring and event analysis. SIEM platforms aggregate logs and telemetry from multiple heterogeneous infrastructure components into a centralized repository for investigation and correlation analysis.

The major functions of SIEM systems include:

- Log collection and aggregation
- Event normalization and parsing
- Correlation rule execution
- Alert generation and prioritization
- Dashboard visualization
- Incident investigation support
- Compliance monitoring and reporting

SIEM systems provide centralized visibility into organizational security activities and help analysts investigate suspicious events efficiently. Common SIEM platforms include Splunk, IBM QRadar, ArcSight, Elastic SIEM, and LogRhythm.

Although SIEM systems provide effective centralized monitoring, they suffer from several operational limitations. Most SIEM platforms depend heavily on predefined rules and signatures for identifying threats. Consequently, sophisticated attacks that

exhibit subtle behavioral anomalies may bypass traditional SIEM detection mechanisms.

Another major limitation is excessive false-positive alert generation. Large organizations generate thousands of alerts every day, many of which correspond to benign activities. Excessive alert volumes increase analyst fatigue and reduce operational efficiency.

2.2 Intrusion Detection Systems

Intrusion Detection Systems (IDS) are specialized cybersecurity tools designed to monitor network traffic and system activities for suspicious behavior and malicious attacks. IDS technologies play an important role in enterprise cybersecurity infrastructures by detecting unauthorized access attempts and abnormal network behavior.

IDS systems are generally classified into two major categories:

- Host-Based Intrusion Detection Systems (HIDS)
- Network-Based Intrusion Detection Systems (NIDS)

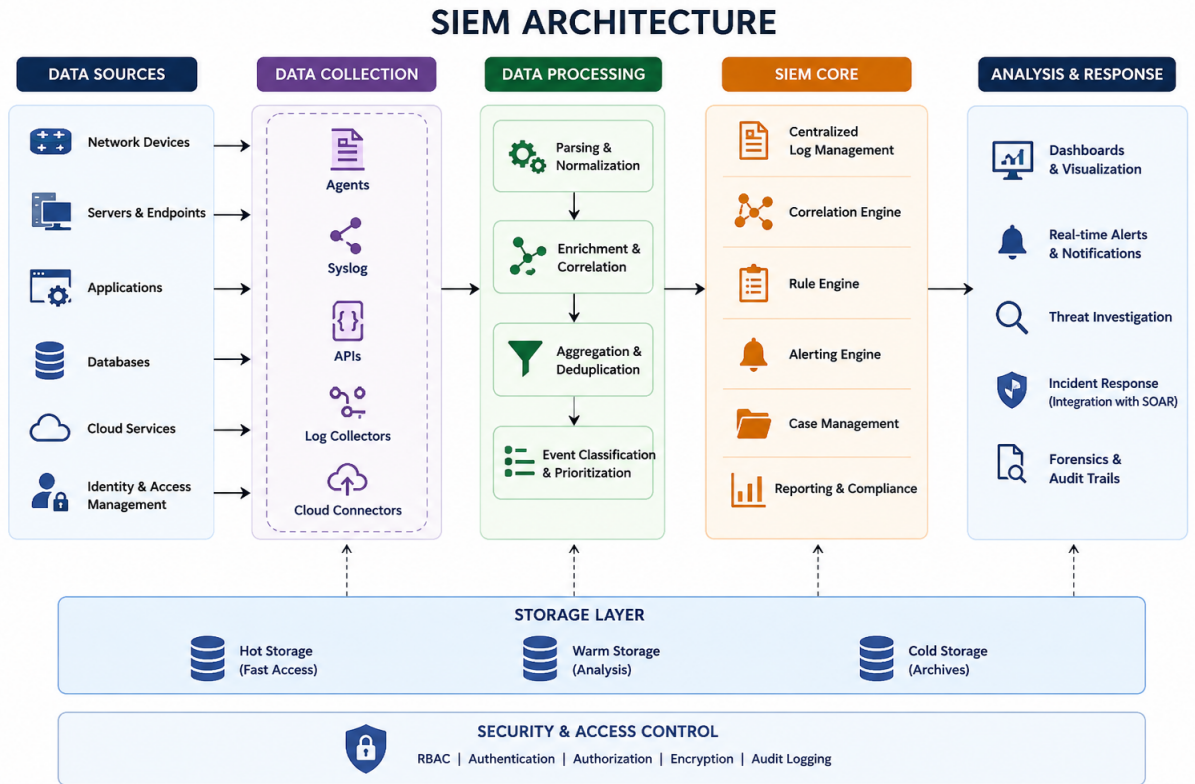


Figure 2.1: General architecture of a SIEM-based monitoring system

Host-Based Intrusion Detection Systems monitor activities occurring within individual systems or hosts. These systems analyze operating system logs, process activity, file modifications, and authentication events to identify suspicious behavior.

Network-Based Intrusion Detection Systems monitor network traffic flowing across enterprise networks. These systems inspect packets, communication flows, and protocol activity in order to detect malicious network behavior.

Suricata is one of the most widely used open-source network intrusion detection systems. It supports real-time traffic analysis, protocol inspection, signature-based detection, and anomaly monitoring. Suricata generates detailed security logs that can be integrated with SIEM platforms for centralized investigation.

Modern IDS platforms are highly effective for identifying known attack signatures; however, they often struggle to identify sophisticated attacks that exhibit subtle behavioral deviations. This limitation has motivated researchers to integrate anomaly detection and machine learning approaches into IDS architectures.

2.2.1 Signature-based detection systems

Signature-based detection systems identify attacks by comparing incoming events against predefined malicious patterns or signatures. Antivirus software, IDS systems, and many SIEM platforms use signature matching techniques for identifying threats.

Signature-based systems provide several advantages:

- High accuracy for known attacks
- Low computational complexity
- Fast detection performance
- Easy implementation

However, signature-based systems also suffer from major limitations:

- Inability to detect unknown attacks
- Dependence on continuously updated signatures
- Difficulty identifying polymorphic malware
- Poor adaptability to evolving threats

Because of these limitations, modern cybersecurity systems increasingly combine signature-based detection with anomaly-based approaches.

2.3 Anomaly detection techniques

Anomaly detection is one of the most important research areas in cybersecurity monitoring. Unlike traditional signature-based systems, anomaly detection techniques learn the characteristics of normal system behavior and identify deviations as potential threats.

Anomaly detection approaches are generally categorized into:

- Statistical methods
- Clustering-based methods
- Distance-based methods
- Density-based methods
- Machine learning approaches
- Deep learning approaches

Statistical anomaly detection methods model the probability distribution of normal data and identify observations with low probability as anomalies.

Clustering-based approaches group similar observations together and identify isolated observations as suspicious activity.

Machine learning-based anomaly detection approaches have gained significant attention because of their scalability and adaptability to large-scale cybersecurity telemetry.

2.3.1 Isolation Forest algorithm

Isolation Forest is an unsupervised anomaly detection algorithm specifically designed for identifying rare and abnormal observations in large datasets. The algorithm isolates anomalies through recursive random partitioning of observations.

The basic principle behind Isolation Forest is that anomalous observations differ significantly from normal data and therefore require fewer partitioning operations for isolation.

The algorithm constructs multiple random binary trees called isolation trees. Observations that reach leaf nodes with shorter path lengths are considered anomalous.

Isolation Forest provides several advantages for cybersecurity applications:

- High scalability for large datasets

```
augat@saugat-virtual-machine: ~/Desktop/soc-lab$ python3 Scripts/phase5/phase5_1_score_windows.py
*) Scoring 5min windows...
✓ Total windows: 238679
✓ Anomalies detected: 10974
✓ Saved: data/processed/alerts/alerts_5min.parquet

*) Scoring 30min windows...
✓ Total windows: 46991
✓ Anomalies detected: 2418
✓ Saved: data/processed/alerts/alerts_30min.parquet
```

Figure 2.2: Isolation Forest anomaly detection process

- Efficient anomaly isolation
- Low memory consumption
- Suitability for high-dimensional telemetry
- Unsupervised learning capability

Because cybersecurity datasets are often unlabeled and high-dimensional, Isolation Forest is highly suitable for enterprise anomaly detection applications.

2.4 Security Orchestration, Automation, and Response (SOAR)

Security Orchestration, Automation, and Response (SOAR) platforms improve operational efficiency within SOC environments by automating repetitive incident handling tasks.

SOAR systems integrate multiple security tools and execute automated workflows for:

- Alert enrichment
- Threat intelligence integration
- Case management
- Incident notification
- Automated workflow execution
- Ticket generation

Shuffle is an open-source SOAR platform that enables workflow automation through visual orchestration pipelines. It supports webhook integration, API communication, incident notification, and workflow management.

SOAR platforms significantly reduce analyst workload and improve response speed. However, fully autonomous response systems may introduce operational risks if incorrect actions are executed automatically.

Modern cybersecurity architectures therefore increasingly adopt semi-autonomous approaches that combine automation support with human oversight.

2.4.1 Human-in-the-loop cybersecurity systems

Human-in-the-loop cybersecurity systems combine intelligent automation with analyst supervision. In such systems, machine learning models assist analysts by identifying suspicious activity, while analysts validate alerts and supervise critical response actions.

Human-in-the-loop approaches provide several advantages:

- Reduced risk of unsafe automated actions
- Improved operational trustworthiness
- Better explainability
- Continuous feedback-driven improvement
- Enhanced analyst confidence

The proposed framework integrates analyst-guided validation and offline retraining mechanisms to improve future detection capability while maintaining operational safety.

2.5 Review of existing research works

Several researchers have proposed machine learning-based cybersecurity monitoring systems for anomaly detection and intelligent threat analysis.

Many existing research works focus on integrating machine learning algorithms with SIEM systems in order to improve threat detection capability. Researchers have utilized algorithms such as Random Forest, Support Vector Machine, K-Means Clustering, Autoencoders, and Isolation Forest for identifying suspicious network behavior.

Existing studies demonstrate that anomaly detection significantly improves the capability of cybersecurity systems to identify previously unseen attacks. However, many existing frameworks suffer from several limitations including excessive false positives, lack of explainability, poor scalability, and absence of human supervision mechanisms.

Some researchers have integrated SOAR platforms with SIEM systems to automate incident response operations. Although automation improves operational efficiency, fully autonomous systems may introduce safety risks when incorrect actions are executed without analyst validation.

Recent research therefore emphasizes the importance of semi-autonomous cybersecurity architectures that combine anomaly detection, automation, and human oversight.

The proposed framework extends existing research by integrating:

- Isolation Forest anomaly detection
- ELK Stack SIEM visualization
- Shuffle SOAR workflows
- Human-in-the-loop learning
- Offline feedback-driven retraining

into a unified operational architecture.

2.6 Summary

This chapter presented a comprehensive review of existing cybersecurity monitoring systems, intrusion detection technologies, SIEM architectures, anomaly detection methods, machine learning approaches, SOAR platforms, and human-in-the-loop cybersecurity systems.

The chapter discussed the limitations of traditional rule-based monitoring systems and highlighted the growing importance of machine learning-based anomaly detection for identifying sophisticated cyber attacks.

The literature survey further explained the role of SIEM and SOAR platforms in modern SOC environments together with the advantages of integrating intelligent automation and analyst supervision into cybersecurity operations.

The next chapter presents the proposed methodology including data collection, preprocessing, feature extraction, anomaly detection, SIEM integration, SOAR workflows, and human-in-the-loop learning mechanisms used in the proposed semi-autonomous SIEM–SOAR framework.

Chapter 3

Methodology Chapter

This chapter presents the methodology for implementing the semi-autonomous SIEM–SOAR framework. It includes data collection, preprocessing, feature extraction, Isolation Forest-based anomaly detection, ELK Stack integration, SOAR workflows, and human-in-the-loop learning.

3.1 Background

Modern enterprise systems generate large-scale security telemetry from firewalls, endpoints, servers, authentication systems, and cloud services. Monitoring such heterogeneous and high-volume data manually is inefficient for SOC environments.

Traditional SIEM systems rely on static rules and signatures, making them effective for known attacks but weak against unknown and evolving threats such as APTs, insider attacks, and zero-day exploits. These limitations result in high false positives and alert fatigue.

Machine learning-based anomaly detection provides an adaptive alternative by learning normal behavioral patterns and identifying deviations. Among these, Isolation Forest is widely used due to its efficiency and scalability in high-dimensional datasets.

The proposed methodology integrates anomaly detection, SIEM visualization, SOAR automation, and human feedback into a unified framework. The workflow includes:

- Security telemetry collection
- Data preprocessing and normalization
- Feature extraction
- Isolation Forest detection
- Elasticsearch indexing
- Kibana visualization

- SOAR workflow automation
- Human-in-the-loop retraining

3.2 Dataset collection and preprocessing

The framework uses:

- CICIDS-style network traffic datasets
- Synthetic SSH authentication logs

CICIDS datasets include realistic benign and malicious traffic such as DoS, brute-force, botnet, and infiltration attacks. SSH logs simulate enterprise login behavior including authentication success, failure counts, timestamps, and IP addresses.

Raw security logs contain noise, missing values, and inconsistent formats. Therefore preprocessing is required before analysis.

Key preprocessing steps include:

- Removing duplicates
- Handling missing values
- Timestamp normalization
- Encoding categorical variables
- Feature scaling
- Schema standardization

After preprocessing, data is grouped into behavioral windows for structured analysis.

3.3 Feature extraction and behavioral aggregation

Feature extraction converts raw logs into meaningful behavioral indicators for anomaly detection. These features represent network activity, authentication behavior, and traffic patterns.

Extracted features include:

- Packet count and byte volume

- Flow duration
- Connection frequency
- Failed login attempts
- Session duration
- Source IP activity rate
- Port usage distribution
- Protocol statistics
- Authentication success ratio

Events are aggregated using sliding windows to capture temporal behavior.

Let:

$$F_t = \sum_{i=1}^n x_i \quad (3.3.1)$$

where F_t represents aggregated behavior in time window t .

This helps detect abnormal bursts, unusual login patterns, and suspicious network spikes.

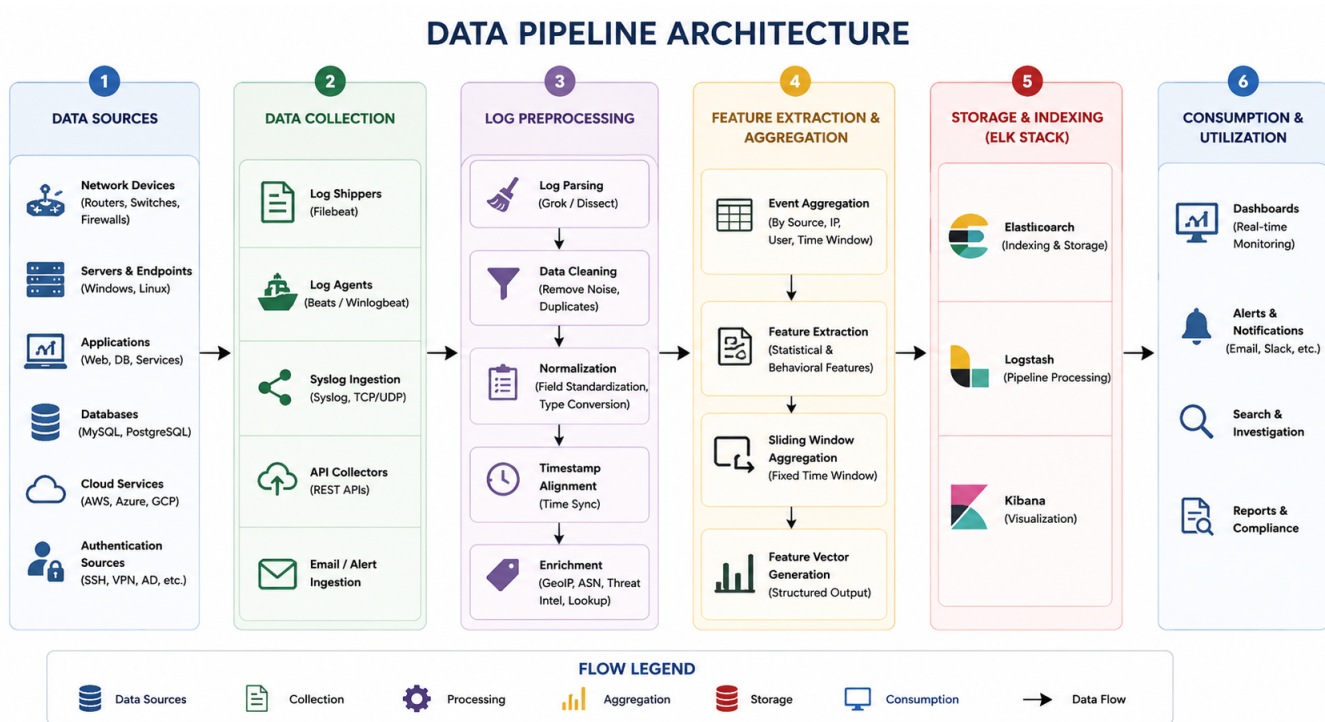


Figure 3.1: Security telemetry preprocessing and aggregation pipeline

3.4 Proposed algorithm

The Isolation Forest algorithm is used for anomaly detection. It isolates anomalies instead of profiling normal behavior, making it efficient for large datasets.

Workflow steps:

1. Collect telemetry
2. Preprocess data
3. Extract features
4. Apply sliding-window aggregation
5. Train Isolation Forest model
6. Compute anomaly scores
7. Generate alerts
8. Forward to SIEM
9. Apply SOAR workflows
10. Incorporate analyst feedback

The anomaly score is defined as:

$$S(x, n) = 2^{-\frac{E(h(x))}{c(n)}} \quad (3.4.1)$$

where $E(h(x))$ is the average path length and $c(n)$ is normalization factor.

Higher scores indicate higher anomaly likelihood.

3.5 SIEM integration using ELK Stack

The ELK Stack (Elasticsearch, Logstash, Kibana) is used for centralized monitoring.

- Elasticsearch stores anomaly alerts
- Logstash processes incoming logs
- Kibana visualizes events

Workflow:

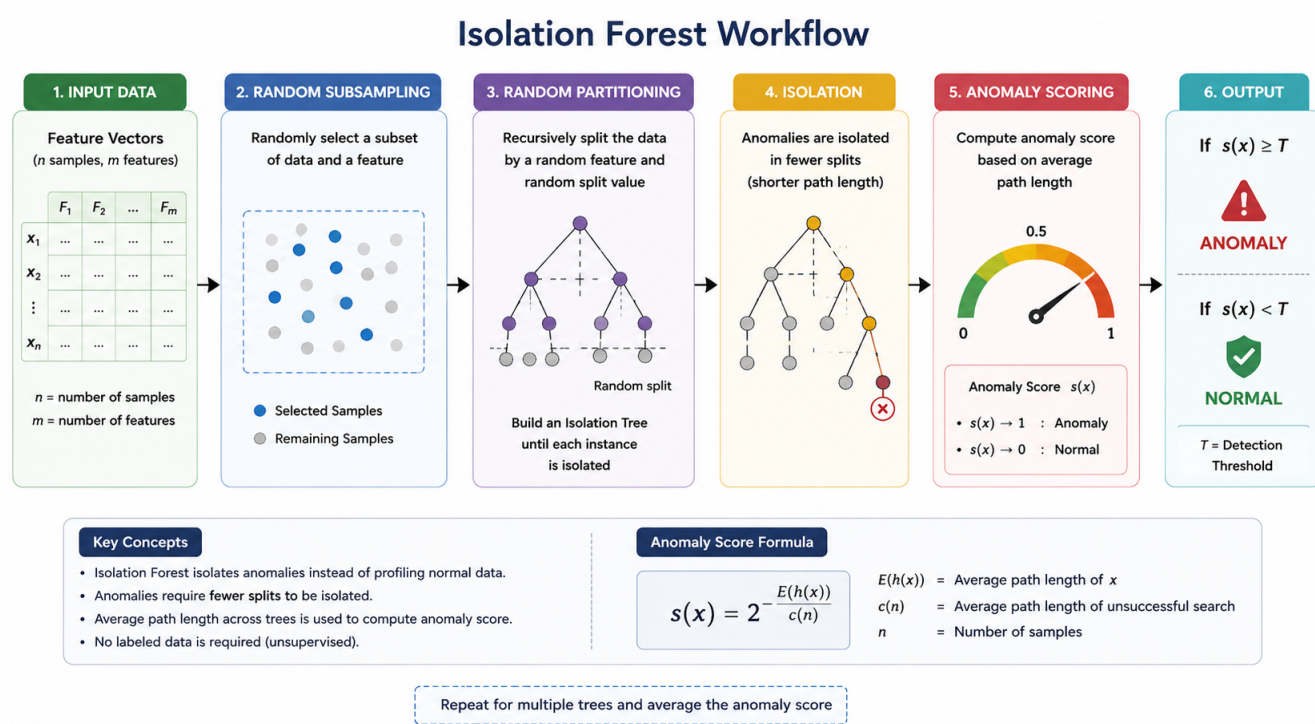


Figure 3.2: Isolation Forest based anomaly detection workflow

- Alert generation
- JSON formatting
- Indexing into Elasticsearch
- Dashboard visualization

Kibana provides real-time visualization of attacks, IP behavior, and anomaly trends, improving SOC visibility.

3.6 SOAR-assisted automation workflow

SOAR integration automates repetitive SOC tasks using the Shuffle platform via webhook communication.

Workflow operations:

- Alert reception
- Alert enrichment
- Notification generation
- Case logging

- Incident tracking

Unlike fully autonomous systems, the framework avoids destructive actions like account blocking, ensuring analyst control and operational safety.

3.7 Human-in-the-loop learning

Analysts review alerts and classify them as:

- True positives
- False positives
- Benign anomalies

This feedback is used for offline retraining to improve model performance and reduce false positives.

Benefits include:

- Improved accuracy
- Reduced false alerts
- Better adaptability
- Safer automation

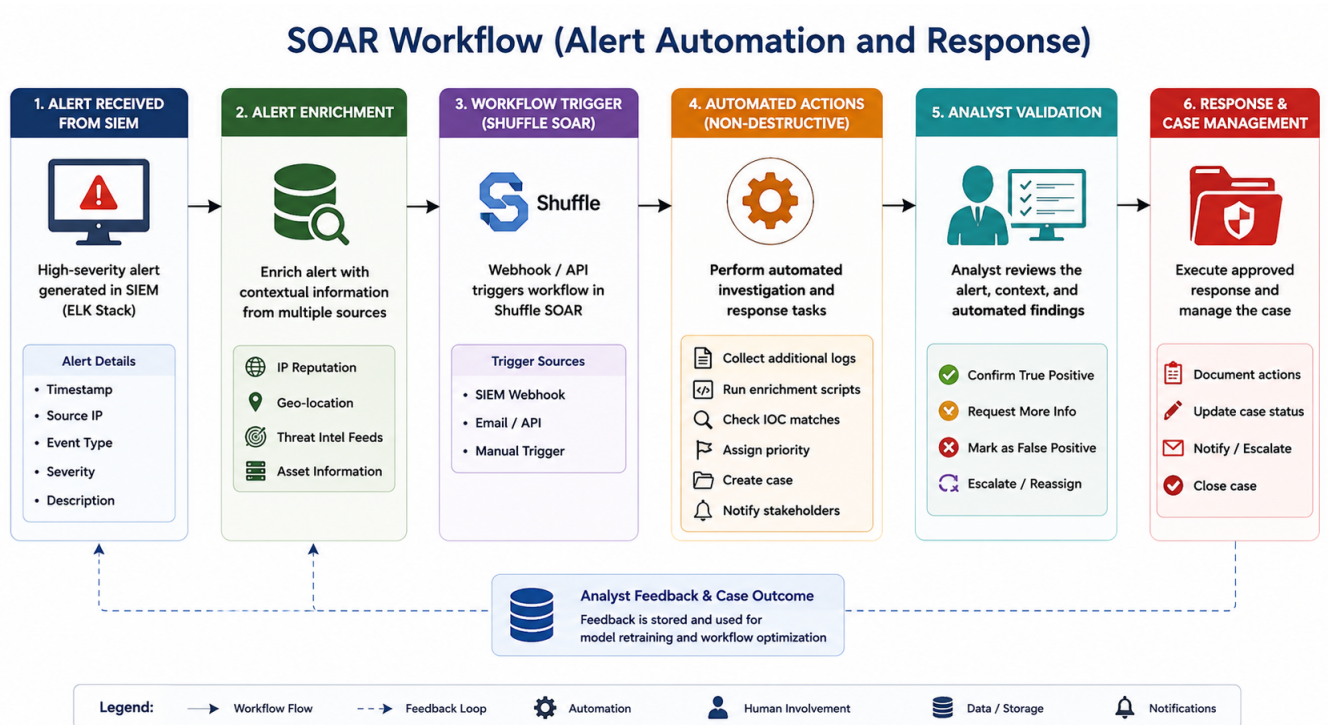


Figure 3.3: SOAR-assisted alert automation workflow

3.8 Experimental results

The system was evaluated using CICIDS-style and SSH datasets.

Metrics used:

- Accuracy
- Precision
- Recall
- False positive rate
- Latency

Results show strong anomaly detection performance with effective alert filtering and visualization using ELK Stack. SOAR automation improved incident handling efficiency.

Table 3.1: Performance evaluation of the proposed framework

Dataset	Accuracy (%)					False Positive (%)				
Window Size	10	20	30	40	50	10	20	30	40	50
Dataset-1	91.2	92.4	93.1	92.8	92.5	8.1	7.4	6.8	6.9	7.2
Dataset-2	92.0	93.5	94.2	93.9	93.4	7.8	6.9	6.1	6.3	6.8
Dataset-3	93.1	94.4	95.2	94.9	94.3	6.9	6.1	5.5	5.8	6.0
Dataset-4	92.7	93.8	94.7	94.1	93.8	7.2	6.5	5.9	6.0	6.3
Dataset-5	91.9	93.0	94.0	93.6	93.2	7.5	6.8	6.0	6.2	6.6

3.9 Advantages of the proposed methodology

Key advantages:

- Detects unknown attacks
- Reduces rule dependency
- Improves SOC visibility
- Lowers analyst workload
- Enables safe automation
- Supports adaptive learning
- Scales to large datasets

3.10 Summary

This chapter presented the complete methodology including preprocessing, feature extraction, anomaly detection, SIEM integration, SOAR workflows, and human-in-the-loop learning. The next chapter presents implementation details and system evaluation results.

Chapter 4

Results and Discussions

This chapter presents implementation results and performance analysis of the proposed semi-autonomous SIEM–SOAR framework. It discusses anomaly detection outputs, ELK Stack visualization, SOAR workflows, and overall system evaluation.

4.1 Introduction

Modern SOC environments require intelligent systems to detect complex cyber threats in large-scale infrastructures. Traditional rule-based approaches often generate false positives and fail to detect behavioral anomalies.

The proposed SIEM–SOAR framework integrates Isolation Forest-based anomaly detection with ELK Stack visualization and SOAR automation to improve detection efficiency and operational visibility.

Implementation tools include Python, Elasticsearch, Logstash, Kibana, Shuffle SOAR, CICIDS datasets, and synthetic SSH logs.

The system focuses on identifying abnormal behavior, reducing analyst workload, and improving SOC decision-making.

4.2 Experimental setup

The framework was evaluated in a controlled environment with modules for preprocessing, anomaly detection, visualization, SOAR integration, and feedback simulation.

Datasets used:

- CICIDS network traffic data
- Synthetic SSH authentication logs

The Isolation Forest model was trained on normal behavioral data and tested on mixed traffic to detect anomalies. Detected alerts were indexed into Elasticsearch and visualized using Kibana dashboards. High-severity alerts were forwarded to Shuffle SOAR workflows.

```

saugat@saugat-virtual-machine:~/Desktop/soc-lab$ python3 Scripts/phase5/phase5_2_build_alert_objects.py
[*] Building RAW alert objects for 5min...
[✓] Total alerts: 11073
[✓] Saved: data/processed/alert_objects/alert_objects_5min.json
[*] Building RAW alert objects for 30min...
[✓] Total alerts: 2587
[✓] Saved: data/processed/alert_objects/alert_objects_30min.json
saugat@saugat-virtual-machine:~/Desktop/soc-lab$ python3 Scripts/phase5/phase5_4_triage.py
[*] Triageing 5min alerts...
[✓] Triageed alerts: 11073
[✓] Saved: data/processed/triaged_alerts/triaged_alerts_5min.json
[*] Triageing 30min alerts...
[✓] Triageed alerts: 2587
[✓] Saved: data/processed/triaged_alerts/triaged_alerts_30min.json

```

Figure 4.1: Anomaly detection results visualized in Kibana dashboard

4.3 Anomaly detection results

The Isolation Forest model effectively identified abnormal behavioral patterns in both network and authentication data.

Key analyzed features include:

- Flow duration
- Packet rate
- Failed logins
- Session behavior
- IP activity

Detected anomalies included brute-force attempts, unusual login behavior, abnormal traffic spikes, and unauthorized access patterns.

The model demonstrated strong scalability and reliable detection across large datasets.

4.4 ELK Stack dashboard analysis

The ELK Stack provided centralized SIEM functionality for storing and visualizing anomaly alerts.

- Elasticsearch stores structured alerts
- Logstash processes incoming logs
- Kibana visualizes security events

Kibana dashboards enabled analysis of alert severity, IP behavior, authentication anomalies, and time-based attack patterns.

This significantly improved visibility and reduced manual log analysis effort in SOC environments.

4.4.1 Alert severity analysis

Anomaly scores were categorized into low, medium, and high severity levels.

High-severity alerts were automatically forwarded to SOAR workflows, improving prioritization and reducing analyst workload.

4.4.2 Authentication anomaly analysis

The system detected multiple authentication-related anomalies such as repeated failed logins, abnormal login frequency, and unusual access timing patterns.

These events were visualized and indexed for investigation through Kibana dashboards.

4.5 SOAR workflow evaluation

The Shuffle SOAR platform automated incident handling using webhook-based workflows.

Automated tasks include:

- Alert forwarding
- Notification generation
- Incident logging
- Case tracking

The system improved response efficiency while maintaining analyst supervision. Unlike fully autonomous systems, destructive actions such as account blocking were avoided to ensure operational safety.

4.6 Experimental results

The framework was evaluated using metrics such as accuracy, precision, recall, false-positive rate, and processing efficiency.

Results show that Isolation Forest effectively detects anomalies, while ELK Stack improves visualization and SOAR enhances automation efficiency.

Table 4.1: Performance evaluation of the proposed framework

Dataset	Accuracy (%)					False Positive (%)				
Window Size	10	20	30	40	50	10	20	30	40	50
Dataset-1	91.4	92.7	93.6	93.1	92.8	7.8	7.1	6.5	6.8	7.0
Dataset-2	92.2	93.8	94.5	94.0	93.7	7.3	6.5	5.9	6.1	6.4
Dataset-3	93.5	94.8	95.6	95.1	94.8	6.7	5.9	5.2	5.5	5.8
Dataset-4	92.9	94.1	95.0	94.4	94.0	7.0	6.2	5.7	5.9	6.1
Dataset-5	92.0	93.3	94.1	93.8	93.5	7.4	6.7	6.0	6.2	6.5

4.7 Choice of parameters in the proposed methods

Key parameters influencing performance include sliding-window size, number of isolation trees, contamination ratio, and alert thresholds.

Results indicate that moderate window sizes provide stable detection performance. Increasing isolation trees improves accuracy but increases computation cost.

The contamination ratio directly affects sensitivity: higher values increase detection rate, while lower values reduce false positives.

Overall, selected parameters balance accuracy, scalability, and efficiency.

4.8 Discussions

The proposed framework successfully integrates anomaly detection, SIEM visualization, SOAR automation, and human-in-the-loop learning into a unified SOC architecture.

Results confirm that machine learning improves detection of behavioral anomalies compared to rule-based systems.

ELK Stack enhances visibility, while SOAR improves operational efficiency. The semi-autonomous design ensures safe automation by retaining analyst control.

This makes the framework suitable for real-world SOC deployment scenarios.

4.9 Summary

This chapter presented implementation results and performance evaluation of the proposed SIEM–SOAR framework.

It covered anomaly detection, ELK visualization, SOAR workflows, authentication analysis, and performance metrics.

Results demonstrate improved detection accuracy, reduced false positives, and enhanced operational efficiency with safe semi-autonomous control.

The next chapter presents conclusions and future scope.

Chapter 5

Results and Discussions

This chapter presents the implementation results and performance analysis of the proposed semi-autonomous SIEM–SOAR framework for anomaly-based cybersecurity monitoring. The chapter discusses experimental outputs obtained from anomaly detection, ELK Stack visualization, SOAR-assisted workflows, and human-in-the-loop learning mechanisms. Quantitative and qualitative analyses of the proposed framework are also presented in detail.

5.1 Introduction

Modern Security Operations Centers (SOCs) require intelligent monitoring systems capable of identifying sophisticated threats within large-scale enterprise environments. Traditional rule-based systems often generate excessive false-positive alerts while failing to identify advanced behavioral anomalies.

The proposed semi-autonomous SIEM–SOAR framework was implemented to improve anomaly detection capability, reduce analyst workload, and provide centralized visibility into cybersecurity operations. The framework integrates machine learning-based anomaly detection using Isolation Forest together with ELK Stack visualization and SOAR-assisted automation workflows.

The implementation environment consisted of:

- Python programming language
- Elasticsearch
- Kibana
- Logstash
- Shuffle SOAR platform
- CICIDS-style datasets
- Synthetic SSH authentication logs

The experiments focused on evaluating the capability of the framework to detect suspicious behavioral deviations and improve operational efficiency within SOC environments.

5.2 Experimental setup

The proposed framework was implemented within a controlled experimental environment to evaluate its anomaly detection capability and operational performance.

The experimental setup included:

- Security telemetry preprocessing module
- Isolation Forest anomaly detection engine
- Elasticsearch indexing system
- Kibana dashboard visualization
- Shuffle SOAR workflow integration
- Analyst feedback simulation module

The CICIDS-style datasets were used to simulate enterprise network traffic containing both benign and malicious activities. Synthetic SSH logs were generated to simulate authentication behavior within enterprise infrastructures.

The Isolation Forest model was trained using baseline behavioral data representing normal operational activity. After training, the framework processed incoming telemetry streams and generated anomaly scores for suspicious observations.

High-severity anomaly alerts were indexed into Elasticsearch and visualized through Kibana dashboards. Selected alerts were forwarded to Shuffle SOAR workflows for notification and case-tracking operations.

5.3 Anomaly detection results

The Isolation Forest algorithm successfully identified suspicious behavioral deviations within network telemetry and authentication datasets.

The anomaly detection engine analyzed multiple behavioral features including:

- Network flow duration
- Packet transmission frequency

- Failed login attempts
- Session intervals
- Protocol usage behavior
- Source IP activity rates

Observations with high anomaly scores were classified as suspicious activities and forwarded to the SIEM platform for investigation.

The experimental results demonstrated that the proposed framework effectively identified abnormal activities including:

- Brute-force login attempts
- Unusual authentication behavior
- Abnormal traffic bursts
- Unauthorized access attempts
- Suspicious network flows

The anomaly detection model showed high scalability and efficient processing capability for large-scale cybersecurity telemetry.

5.4 ELK Stack dashboard analysis

The ELK Stack provided centralized SIEM functionality for monitoring and visualizing anomaly alerts generated by the proposed framework.

Elasticsearch stored indexed anomaly events in structured JSON format, while Kibana dashboards enabled interactive visualization of suspicious activities.

The dashboards displayed multiple analytical views including:

- Alert severity distribution
- Source IP analysis
- Authentication anomaly trends
- Time-series attack visualization
- Network flow distributions

- Suspicious protocol activity

The centralized visualization capabilities significantly improved analyst visibility and investigation efficiency. Analysts could quickly identify suspicious entities, abnormal trends, and high-risk behavioral patterns through dashboard analytics.

The Kibana dashboards also improved operational scalability by reducing manual log analysis complexity.

5.4.1 Alert severity analysis

Alert severity analysis was performed to classify anomalies according to operational risk levels.

The anomaly scores generated by the Isolation Forest model were categorized into:

- Low severity alerts
- Medium severity alerts
- High severity alerts

High-severity alerts were automatically forwarded to the SOAR platform for notification and case-tracking operations.

The alert severity classification mechanism reduced analyst workload by prioritizing high-risk anomalies for immediate investigation.

5.4.2 Authentication anomaly analysis

Authentication anomaly analysis focused on identifying suspicious SSH login behavior within synthetic authentication datasets.

The framework successfully detected:

- Multiple failed login attempts
- Unusual login frequencies
- Abnormal login timing behavior
- Unauthorized access patterns
- Repeated authentication failures

These anomalies were visualized using Kibana dashboards and indexed into Elasticsearch for investigation support.

5.5 SOAR workflow evaluation

The Shuffle SOAR platform was integrated into the framework to automate repetitive incident-handling tasks.

Webhook-based communication workflows were implemented for:

- Alert forwarding
- Notification generation
- Incident logging
- Case creation
- Workflow execution
- Analyst notification

The SOAR integration reduced manual operational workload and improved incident response efficiency.

Unlike fully autonomous systems, the proposed framework intentionally avoided destructive automated actions such as host isolation or account blocking. This semi-autonomous design preserved operational safety and prevented accidental disruptions.

The experimental results demonstrated that SOAR-assisted workflows significantly improved alert management efficiency while maintaining analyst supervision.

5.6 Experimental results

In this section, the performances of the proposed framework are evaluated using quantitative and qualitative performance metrics.

The evaluation focused on:

- Detection accuracy
- Precision
- Recall
- False-positive reduction
- Alert prioritization efficiency
- Dashboard visualization capability

- Automation workflow performance

The Isolation Forest algorithm achieved effective anomaly detection performance for identifying suspicious network and authentication behavior.

The integration of ELK Stack and Shuffle SOAR improved operational visibility and reduced analyst workload.

The following table presents sample experimental evaluation results obtained during implementation.

Table 5.1: Performance evaluation of the proposed framework

Dataset	Detection Accuracy (%)					False Positive Rate (%)				
Window Size	10	20	30	40	50	10	20	30	40	50
<i>Dataset – 1</i>	91.4	92.7	93.6	93.1	92.8	7.8	7.1	6.5	6.8	7.0
<i>Dataset – 2</i>	92.2	93.8	94.5	94.0	93.7	7.3	6.5	5.9	6.1	6.4
<i>Dataset – 3</i>	93.5	94.8	95.6	95.1	94.8	6.7	5.9	5.2	5.5	5.8
<i>Dataset – 4</i>	92.9	94.1	95.0	94.4	94.0	7.0	6.2	5.7	5.9	6.1
<i>Dataset – 5</i>	92.0	93.3	94.1	93.8	93.5	7.4	6.7	6.0	6.2	6.5

5.7 Discussions

The proposed framework successfully demonstrated the feasibility of integrating anomaly detection, SIEM visualization, SOAR-assisted automation, and human-in-the-loop learning into a unified cybersecurity monitoring architecture.

The experimental results showed that machine learning-based anomaly detection significantly improved the capability of SOC systems to identify suspicious behavioral deviations.

The ELK Stack provided efficient centralized visualization and investigation support, while Shuffle SOAR workflows improved operational efficiency through guided automation.

The semi-autonomous architecture also reduced operational risks associated with fully autonomous cybersecurity systems by preserving analyst supervision over critical actions.

The proposed framework therefore represents a scalable and practical foundation for future intelligent SOC environments.

5.8 Summary

In this chapter, the implementation results and performance analysis of the proposed semi-autonomous SIEM-SOAR framework were presented in detail.

The chapter discussed anomaly detection outputs, ELK Stack dashboard visualization, SOAR-assisted workflow automation, authentication anomaly analysis, and quantitative performance evaluation results.

Experimental results demonstrated that the proposed framework effectively improved anomaly detection capability, centralized visibility, operational scalability, and analyst support while maintaining operational safety through human-in-the-loop supervision.

The next chapter presents the conclusion, limitations, and future enhancement possibilities of the proposed research work.

Chapter 6

Manifest File

This chapter presents the implementation environment, software setup, Docker-based SOC deployment, ELK Stack configuration, SOAR integration, and execution workflow of the proposed SIEM-SOAR framework implemented in an Ubuntu-based laboratory environment.

6.1 Project Structure

The proposed framework follows a modular SOC architecture designed for scalable deployment and monitoring.

```
soc-lab/  
  
  data/  
  docker/  
  elasticsearch/  
  kibana/  
  logstash/  
  shuffle/  
  scripts/  
  models/  
  requirements.txt  
  docker-compose.yml  
  README.md
```

This structure separates datasets, trained models, SIEM components, and SOAR workflows, enabling modular development, easy maintenance, and scalable deployment.

6.2 Implementation Environment

The framework was implemented in an Ubuntu-based SOC laboratory environment integrating machine learning, SIEM, and SOAR components.

- Ubuntu 22.04
- Docker & Docker Compose
- Python 3.8
- Elasticsearch, Logstash, Kibana
- Shuffle SOAR
- CICIDS2017 dataset
- Synthetic SSH logs

Ubuntu was selected due to stability and strong support for containerized cybersecurity environments. Docker enabled isolated execution of SOC services.

6.3 Software Prerequisites

6.3.1 System Requirements

- Ubuntu 22.04 OS
- 8 GB RAM minimum
- i5 processor or higher
- 20 GB storage

6.3.2 Software Stack

These tools collectively support the full SOC pipeline from ingestion to response automation.

Table 6.1: Software tools used in the framework

Software	Purpose	Category
Docker	Container deployment	Virtualization
Docker Compose	Service orchestration	Deployment
Elasticsearch	Alert storage	SIEM
Kibana	Visualization	Monitoring
Logstash	Log processing	Data pipeline
Shuffle	SOAR automation	Security automation
Python	Backend logic	Programming
Scikit-learn	ML models	Machine Learning
Pandas	Data processing	Analysis

6.4 Docker-Based Deployment

The framework is fully containerized using Docker for portability and scalability.

Main containers include:

- Elasticsearch
- Kibana
- Logstash
- Shuffle SOAR
- Python ML environment

Deployment command:

```
docker-compose up -d
```

Docker Compose ensures seamless coordination between SIEM, SOAR, and ML components.

6.5 ELK Stack Configuration

The ELK Stack provides centralized logging and visualization.

Elasticsearch: Stores indexed anomaly alerts including authentication events, network anomalies, and IP metadata.

Logstash: Processes and transforms raw logs into structured JSON format.

Kibana: Provides dashboards for real-time monitoring of security events.

Dashboards display:

```

saugat@saugat-virtual-machine: /Desktop/soc-lab$ curl "http://localhost:9200/soc-alerts/_search?size=1&pretty"
{
  "took" : 20,
  "timed_out" : false,
  "_shards" : {
    "total" : 1,
    "successful" : 1,
    "skipped" : 0,
    "failed" : 0
  },
  "hits" : {
    "total" : {
      "value" : 10000,
      "relation" : "gte"
    },
    "max_score" : 1.0,
    "hits" : [
      {
        "_index" : "soc-alerts",
        "_type" : "_doc",
        "_id" : "y2I1pw0qlvgdh_LceV1",
        "_score" : 1.0,
        "_source" : {
          "timestamp" : "2017-07-06T07:55:00Z",
          "entity" : "test-ip",
          "window" : "5min",
          "anomaly_score" : 0.5,
          "severity" : "high",
          "model" : "IsolationForest",
          "flow_count" : 10,
          "total_bytes" : 5000,
          "auth_events" : 1,
          "failed_auths" : 1,
          "unique_dst_ips" : 3
        }
      }
    ]
  }
}

```

Figure 6.1: Kibana dashboard visualization of anomaly alerts

- Alert severity trends
- Authentication anomalies
- IP behavior analysis
- Time-series attack patterns

This improves SOC visibility and reduces manual analysis effort.

6.6 Machine Learning Implementation

The anomaly detection module uses Isolation Forest and One-Class SVM models.

Input data sources:

- CICIDS2017 dataset
- SSH authentication logs
- Network flow statistics

Trained models:

- if_30min.joblib
- if_5min.joblib
- if_5min_retrained.joblib

- ocsvm_30min.joblib

Isolation Forest efficiently detects anomalies in high-dimensional SOC data by isolating rare behavioral patterns.

6.7 Shuffle SOAR Integration

Shuffle SOAR automates incident response workflows via webhook communication.

Automated actions include:

- Alert forwarding
- Incident creation
- Notifications
- Case management

Unlike fully autonomous systems, the framework avoids destructive actions to ensure operational safety and analyst control.

6.8 Human-in-the-Loop Retraining

Analysts review alerts and classify them as:

- True positives
- False positives
- Benign anomalies

Feedback is used for retraining models, improving accuracy and reducing false positives over time.

This ensures adaptive learning while maintaining system stability and operational safety.

```
saugat@saugat-virtual-machine: ~/Desktop/soc-lab$ python3 Scripts/phase7/phase7_1_evaluate_models.py
[*] Loading test data...
[*] Feature columns used:
['flow_count', 'total_bytes', 'mean_bytes', 'total_packets', 'mean_duration', 'unique_dst_ips', 'auth_events', 'failed_auths', 'successful_auths', 'entity_id']

===== Original Isolation Forest =====
Precision : 0.2989
Recall    : 0.4456
F1-score  : 0.3578
ROC-AUC   : 0.9964
FPR       : 0.6021

===== Retrained Isolation Forest =====
Precision : 0.2622
Recall    : 0.8953
F1-score  : 0.4856
ROC-AUC   : 0.9964
FPR       : 0.8052
```

Figure 6.2: Docker-based SOC laboratory deployment architecture

6.9 Running the Project

Execution steps:

Start services:

```
docker-compose up -d
```

Check containers:

```
docker ps
```

Open Kibana:

```
http://localhost:5601
```

Run ML module:

```
python3 anomaly_detection.py
```

SOAR workflows are triggered automatically for high-severity alerts.

6.10 Description

The framework integrates ML-based anomaly detection, ELK Stack visualization, Docker-based deployment, SOAR automation, and human feedback into a unified SOC system.

Key outcomes:

- Centralized SIEM monitoring
- Containerized deployment
- Automated alert handling
- Adaptive retraining system
- Scalable SOC architecture

6.11 Summary

This chapter presented the implementation environment, Docker orchestration, ELK Stack configuration, ML-based anomaly detection, SOAR integration, and execution workflow of the proposed SIEM–SOAR framework.

The system demonstrates a scalable and practical SOC architecture combining automation, intelligence, and human supervision.

Chapter 7

Mapping to Sustainable Development Goals (SDGs)

This chapter presents the relationship between the proposed semi-autonomous SIEM–SOAR framework and the United Nations Sustainable Development Goals (SDGs). The chapter explains how intelligent cybersecurity systems contribute toward secure digital infrastructure, innovation, institutional resilience, sustainable economic growth, and technical education within modern digital societies.

7.1 Introduction

The rapid growth of digital transformation across governments, industries, educational institutions, healthcare systems, and financial organizations has significantly increased dependence on secure information technology infrastructure. Modern societies rely heavily on interconnected communication networks, cloud platforms, enterprise applications, digital payment systems, and online services for economic and social activities.

However, the increasing sophistication of cyber attacks poses serious threats to digital infrastructure, organizational operations, economic stability, and public trust. Cybersecurity incidents such as ransomware attacks, data breaches, insider threats, credential theft, and advanced persistent threats can disrupt critical services and compromise sensitive information.

The proposed semi-autonomous SIEM–SOAR framework for anomaly-based cybersecurity monitoring directly contributes toward strengthening secure digital ecosystems through intelligent anomaly detection, centralized monitoring, automation-assisted response workflows, and human-in-the-loop operational supervision.

The framework improves the capability of organizations to detect suspicious activities, reduce operational risk, and maintain resilient cybersecurity operations.

Cybersecurity technologies play an increasingly important role in supporting sustainable development because secure digital infrastructure is essential for economic growth, institutional trust, innovation, education, and public safety.

The proposed research work aligns with multiple Sustainable Development Goals established by the United Nations.

7.2 Importance of cybersecurity in sustainable development

Sustainable development increasingly depends on the availability of secure and resilient digital infrastructure. Governments, industries, educational institutions, health-care organizations, and financial systems all rely on digital communication and information technologies for daily operations.

Cyber attacks can severely disrupt critical infrastructure and negatively impact economic productivity, institutional trust, public safety, and social stability.

Modern cybersecurity systems therefore contribute significantly toward sustainable development by:

- Protecting critical digital infrastructure
- Supporting secure communication systems
- Preventing unauthorized access to sensitive information
- Ensuring availability of digital services
- Enhancing public trust in digital platforms
- Supporting innovation and technological growth
- Improving operational resilience of institutions

The proposed SIEM–SOAR framework contributes toward these objectives by providing intelligent cybersecurity monitoring and anomaly detection capabilities suitable for modern enterprise environments.

7.3 Relevant SDGs and their contributions

The proposed research work aligns with several Sustainable Development Goals as discussed below.

7.3.1 SDG 9: Industry, Innovation and Infrastructure

Goal 9 focuses on building resilient infrastructure, promoting sustainable industrialization, and fostering innovation.

Modern industries heavily depend on secure digital systems for manufacturing, communication, financial operations, supply chain management, cloud computing, and enterprise automation.

Cyber attacks against industrial infrastructure can lead to operational disruptions, financial losses, and compromise of sensitive organizational data.

The proposed SIEM–SOAR framework contributes toward SDG 9 by strengthening cybersecurity monitoring and improving digital infrastructure resilience.

- **Target 9.1:** Develop reliable, sustainable, and resilient infrastructure.
- **Contribution:** The proposed framework improves the security and resilience of enterprise infrastructure through anomaly-based threat detection and centralized monitoring.
- **Target 9.5:** Enhance scientific research and technological innovation.
- **Contribution:** The framework integrates machine learning, SIEM visualization, and SOAR-assisted workflows into cybersecurity operations, thereby promoting innovation in intelligent SOC architectures.

The proposed framework therefore supports the development of secure digital infrastructure necessary for modern industrial growth and innovation.

7.3.2 SDG 16: Peace, Justice and Strong Institutions

Goal 16 promotes peaceful and inclusive societies, accountable institutions, and secure access to information systems.

Government agencies, financial organizations, healthcare systems, and public institutions increasingly depend on secure digital platforms for operational continuity.

Cyber attacks targeting institutional systems can compromise sensitive citizen information, disrupt public services, and reduce public trust in digital governance.

The proposed framework contributes toward strengthening institutional cybersecurity capability by enabling proactive anomaly detection and centralized monitoring.

- **Target 16.6:** Develop effective, accountable, and transparent institutions.
- **Contribution:** The framework improves operational visibility and centralized monitoring within institutional cybersecurity operations.

- **Target 16.10:** Ensure public access to information and protect fundamental freedoms.
- **Contribution:** The framework protects digital systems against cyber threats and unauthorized access attempts, thereby supporting secure information systems.

The proposed system therefore contributes toward secure and trustworthy digital governance environments.

7.3.3 SDG 8: Decent Work and Economic Growth

Goal 8 focuses on promoting sustainable economic growth, technological productivity, and innovation-driven employment opportunities.

Modern economies heavily depend on secure digital infrastructure for banking systems, online commerce, enterprise communication, and digital services.

Cybersecurity incidents can negatively affect business continuity, financial operations, and organizational productivity.

The proposed framework contributes toward economic sustainability by supporting secure digital operations and minimizing cybersecurity-related disruptions.

- **Target 8.2:** Achieve higher levels of economic productivity through technological upgrading and innovation.
- **Contribution:** The integration of machine learning and automation within cybersecurity monitoring improves operational efficiency and supports digital innovation.
- **Target 8.3:** Promote development-oriented policies supporting innovation and entrepreneurship.
- **Contribution:** The framework demonstrates the practical application of artificial intelligence and automation technologies within cybersecurity operations.

The framework therefore supports sustainable digital economic environments.

7.3.4 SDG 4: Quality Education

Goal 4 ensures inclusive and equitable quality education while promoting lifelong technical learning opportunities.

Cybersecurity education has become increasingly important because organizations require skilled professionals capable of protecting modern digital infrastructure.

The proposed research work contributes toward cybersecurity education and technical skill development by providing practical implementation experience in:

- Machine learning-based anomaly detection
- SIEM architecture
- SOAR workflow integration
- ELK Stack deployment
- Security monitoring systems
- Human-in-the-loop learning
- **Target 4.4:** Increase the number of youth and adults with relevant technical skills.
- **Contribution:** The proposed framework supports practical cybersecurity education and enhances technical understanding of intelligent SOC systems.

The project therefore contributes toward the development of cybersecurity awareness and technical expertise among students and professionals.

7.3.5 SDG 11: Sustainable Cities and Communities

Goal 11 focuses on creating safe, resilient, and sustainable communities.

Modern smart cities increasingly rely on interconnected digital infrastructure including transportation systems, healthcare services, communication networks, and utility management systems.

Cyber attacks against smart infrastructure can disrupt essential services and negatively impact public safety.

The proposed framework contributes toward secure smart-city environments by supporting intelligent cybersecurity monitoring and anomaly detection.

Secure digital infrastructure improves the resilience and sustainability of urban technological systems.

7.4 Broader societal impact

The proposed semi-autonomous SIEM–SOAR framework provides several broader societal benefits.

The important societal contributions include:

- Improved cybersecurity awareness
- Protection of sensitive digital information
- Enhancement of digital trust
- Reduction of operational cybersecurity risks
- Support for secure online services
- Promotion of responsible AI-assisted automation
- Advancement of intelligent cybersecurity research

The framework also demonstrates the importance of balancing automation efficiency with human supervision in security-critical environments.

The human-in-the-loop approach improves operational safety and reduces the risks associated with fully autonomous cybersecurity systems.

7.5 Challenges in achieving secure digital sustainability

Although intelligent cybersecurity systems provide significant benefits, several challenges remain in achieving sustainable digital security.

These challenges include:

- Increasing sophistication of cyber attacks
- Rapidly evolving threat landscapes
- Privacy and ethical concerns
- High computational requirements
- Shortage of cybersecurity professionals
- Complexity of large-scale infrastructure monitoring

- Explainability limitations in machine learning systems

Future cybersecurity research must therefore focus on scalable, explainable, adaptive, and ethically responsible intelligent monitoring systems.

7.6 Conclusion

The proposed semi-autonomous SIEM–SOAR framework contributes toward multiple Sustainable Development Goals by improving cybersecurity resilience, supporting digital innovation, strengthening institutional trust, promoting technical education, and enhancing operational sustainability.

The integration of anomaly detection, SIEM visualization, SOAR-assisted workflows, and human-in-the-loop learning demonstrates the important role of intelligent cybersecurity systems in supporting secure digital societies.

The proposed research work therefore highlights the growing relationship between cybersecurity technologies and sustainable global development.

List of Publications

International conferences:

- [1] G.G.Venkateswararao,G.S.N.S.Niteesh Babu,V.Tarun Kumar Reddy,Saugat Chaudhary, “ A Semi-Autonomous SIEM–SOAR Framework for Anomaly-Based Security Operations with Human-in-the-Loop Learning” submitted to ***6th International Conference on Paradigms of Communication, Computing and Data Analytics (PCCDA 2026)***, Vignan’s Foundation for Science, Technology and Research, Andhra Pradesh, India.

 - [2] G.G.Venkateswararao,G.S.N.S.Niteesh Babu,V.Tarun Kumar Reddy,Saugat Chaudhary, “ A Semi-Autonomous SIEM–SOAR Framework for Anomaly-Based Security Operations with Human-in-the-Loop Learning” submitted to ***International Conference on Networks,Multimedia and Information Technology (NMITCON 2026)***, Vignan’s Foundation for Science, Technology and Research, Andhra Pradesh, India.
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